

ActInf Livestream #032.0 ~ "Stochastic Chaos and Markov Blankets"

Livestream | 1:26:33

hello and welcome everyone to actin flab

live stream number 32.0

at the actin flab the paper we're

discussing is going to be stochastic

chaos and markov blankets and it's

november 7th

2021

let's play our theme

[Music]

welcome to the actinf lab everyone we

are a participatory online lab that is

communicating learning and practicing

applied active inference

you can find us at the links on this

page

this is a recorded and an archive live

stream so please provide us with

feedback so that we can improve on our
work

all backgrounds and perspectives are

welcome here and we'll be following

video etiquette for live streams

at this short link you can see our past

and upcoming live streams and here on

the main tab the live stream calendar

for 2021

we're here in early november we had our

fourth quarterly lab round table

summarizing the kinds of projects that

you can get involved in

and what we had done in the previous

year what we're looking forward to next

year and then the first two weeks of

discussion are going to be on november

9th and 16th for this paper that will be

discussed tonight

in

stream number 33 we'll have abel's paper
thinking like a state and we haven't set
the papers yet for 34 and 35.

the goal of

32.0 is to learn and discuss this paper
stochastic chaos and markov's blankets
2021

by carl firsten connor hines kai

oldhofer lancelet decosta and thomas

parr and just like all the dot zeros

this is just an introduction to some of

the ideas it's not a review or a final

word and especially if you have any

experience

or you want to learn more about any of

these areas and help improve our

presentation or

understanding of it that'd be very

helpful you can join a live stream or

just come get involved in some of the

aspects of the lab because

these are things that we want to

understand but also there's a lot to

learn

and in the coming two weeks we're going

to discuss this paper

so i'm daniel and i'm in california

the big questions of this paper

are related to the general

discussion in the active inference and

free energy principle literature and

beyond what is a good model of thingness

in a chaotic dynamic and dissipative

world so do we think of things as as

they are in the snapshot or do we have

to think of them through time at what

spatial or temporal scale or with what
kind of measure stick does it make sense
to talk about things
and then how does the result of that
first question this model or approach to
thickness speak to
system sentence which is going to be a
word that is used in this paper
and
notably this is not being defined in
terms of feeling like experiencing
it's
defined in the paper as where internal
states look as if they're inferring
external states so perhaps closer to
what dennett would call the intentional
stance
but this is sort of just pragmatism
meets
anti-dissipativism
third point what is a markov blanket and
how is one modeled and identified or
defined statistically this is going to
be
the technical
bulk and contribution
of the paper so
if
anyone wants to learn more and help us
present some of the technical details
which we're going to go through in this
lecture but total disclaimer if it's not
accurate or it's an incorrect
generalization
a word that wasn't a key word per se but
maybe is just a qualitative entry point
is this idea of flow and here we have
people flowing like a

long time exposure in a subway station
and there's water flow which has several
aspects the water itself is moving but
also there's the flow of energy
there's information flow and then
there's this psychological concept of
flow

and so

different aspects of these

could be said to be flowing in different
ways

and people jump across these different
areas so wouldn't it be cool if there
were a quantitative model that kind of
incorporated some of what was meant
by all these concepts including
a science of

perhaps perception cognition in action
so the big picture is how are we going
to connect kind of physical
flow

models

to potentially other kinds

like macro systems and informational and
even psychological or behavioral
paper is stochastic chaos and markov
blankets

2021

and uh

two of the aims and claims of the paper
of which there are definitely multiple

so this is just a few of them

in this work we attempt an end-to-end
derivation of a markov blanket starting
with a normal form of stochastic chaos
and ending with a functional form of a
synchronization map

the second

aim

or like a summary of the paper is we
conclude with the discussion of the
implications and generalizations of the
following heuristic

so not necessarily formal but more like
using empirical simulation

or approximation proof of principle for
understanding simulating and
characterizing an elemental form of
sentience in systems that self-organize
to non-equilibria

so that is where sentience comes in and
again this isn't

saying that because we did some matrix
math

we think it has phenomenological
awareness sentience is going to be more
like intentional stance

it's resisting dissipation perhaps it's
acting strategically given the
affordances in its niche

so one really relevant paper
for stochastic chaos and markov blankets
is

the paper bayesian mechanics for
stationary processes with multiple of
the same authors

and um this one was put on archive on
june 25th 2021

and then this one looks like it was
received by the journal

at

july 9th 2021 so

one

question is like to to what extent are
these complementary
are these requisite or are they

somehow related to each other or are
they two independent
kind of dimensions of advance
there's some overlapping areas but also
there's some different areas so that
would be good to
learn about
so there's the keywords of the paper
were bayesian thermodynamics
information geometry
variational inference and markov blanket
so let's go just
one level deeper into the flow example
we have a lot of
fields like
math physics information computing
thermodynamics geometry
a few other fields
where there are models of physical flow
so
everything from heat engines and heat
flow to
quantum
and classical computation
clouds and everything else so
this is a engineering experience
and ontology
and um from multiple different areas of
mathematical formalisms that sometimes
map very closely to the real world or
not
then
there's these again flow
models
or process models for mental flow for
example thinking about attentional
allocation or informational processing
and

the big idea which is proposed

in

fristen at all 2006 at least one

phrasing of it in an image that gets

referenced back many times is this idea

of like a winged snowflake

and so

below the phase boundary so above

the temperature of melting ice

the droplets are unable to

enact policy so they simply

fall from phase

and then they become rain

but above the solid phase boundary

so

below the freezing point of water

there are the snowflake structures so

there's an organization

and then

one can imagine that if this shape above

the phase boundary

is able to enact action on the

environment

first if nothing other than

by chance

you would find that there'd be for

example persistence of the most adaptive

action policies

so that is the exchange

and the interfacing of active systems

with their environment that precludes

the phase transition so that's the

winged snowflake example

of

early er fristen

and it will come back at the end in a

pretty interesting way

so one of the first keywords is bayesian

here's rev bays and we've talked a lot
about bayesian things so just to kind of
pick a few new images
here's some overlapping
between frequentism
here
and bayesianism here and
some of the overlap so it's just kind of
interesting to look at
and what exactly does it mean for a
theory or for statistics to be bayesian
explicit modeling of priors
and here's the sort of classic
bayes theorem which
i'm just not going to go into right now
i'm sure there's awesome resources to
learn more about bayesian statistics
and just sort of the way that it becomes
meaningful is just embodied how we want
to be able to bring our full knowledge
to bear
on scientific questions or any other
kind of question we want to be able to
integrate different kinds of data sets
and have models that can be able to
receive data as well as generate it
these things are all tractable in modern
bayesian
frameworks
thermodynamics is the quoting wikipedia
here branch of physics that deals with
heat work and temperature and the
relation to energy radiation and
physical properties of matter
so there's this sort of thermodynamic
patterns or laws in the way that they
relate to statistical mechanics so how
the kind of jostling of molecules

leads to the ideal gas law

or

kind of work

capacity in

temperature

engines

and then just looking around there was

kind of just some old ideas some new

ideas

so this is something we've talked about

before with

decision making but also

it's um

how we think about chemical reactions in

terms of

the lower activation energy is the

kinetic product that's like the one that

is easily jostled to

and then the one that has a higher

uh activation energy is not favored to

occur but then it sometimes can be like

a lower thermodynamic product so

releasing more ΔG

and then

if we're going to have some sort of an

information thermo synthesis

what does

ΔG

informationally look like like what is

the informational

imperative

the title of the paper has stochastic

chaos

so just

found a few

different

citations on it

and it's an idea that

has been going back since before 2000s
so i don't know when the exact beginning
was but it's been around so that was
just one
question for uh the authors or for
anyone who's knowledgeable about it what
is uh the scope of stochastic chaos and
how
how do we
think about the relative
importance or meaning of
solving one category
of
dynamic systems
so what is stochastic chaos
stochastic
subject to probabilistic variation in
one way or another and we'll kind of
like look at multiple ways that that can
come into the picture through sampling
or approximation and then chaos which is
sensitivity to small changes which is
summarized
at least in one way by the lyapunov
exponent which is a measure of how
closely
separated points either converge or
diverge
as a feature of the shape
and the type of the matrices that
describe the systems
another keyword was information geometry
so
i'm definitely not the person to
give a specific definition
it would be great to learn more from
someone who could link this to the
specific

modern advances so a few people who have
emailed us

but uh john baez who is a professor in
the uc system
wrote

on
his site which has a whole curriculum on
information geometry

not that i've taken it but heard it was
good

information geometry is the study of
statistical manifolds which are spaces
where each point is a hypothesis about
some state of affairs

this subject usually considered a branch
of statistics has important applications
to machine learning and somewhat
unexpected connections to evolutionary
biology

so

this image

it just kind of was like if these are
dimensions of

something abstract

that projected down onto the graph paper

is going to look like a triangle

and then this shape which maybe is kind
of like a saddle shape or if the

two um distal parts are connected it's
like a tetrahedra in a way

then

that projected down from a specific
angle is going to look like a

right angled

square

but from a different angle it would have
a different profile

so the same tetrahedra

rotated
in different ways
can project down into a different
geometry
so
kind of interesting to think about that
with
information and if you're measuring
several axes of information you have an
informational tetrahedra
as if it doesn't need to be
saying that it is actually that way
one of the other keywords was
variational bayesian inference and this
was from the slides of 26
and some also good blog posts that
others have written
it
was defined in this blog post um by
variational inference methods that
consist uh consist in finding the best
approximation of a distribution
among a parameterized family so it's
kind of like finding the best linear
model for a data set this is a way to
fit more
complex
data
and models but it keeps it constrained
within a family so that you're just
tuning the knobs that you know
interact in a good way
versus all the possible degrees of
freedom in modeling which is basically
infinite
and then
two ways that bayesian approaches can
be deployed there's probably many more

than this too

there's the sampling based approach or

the monte carlo

like

gambling approach which is like

pointillism where it kind of samples and

then this if the underlying distribution

you're kind of agnostic to it if you run

this properly and the underlying

distribution has the right properties

you can sample something that's totally

adequate for making good decisions

and then the alternate approach is kind

of like the svg approach the variational

inference you kind of have a

library of cat or of mammal subunit

vectors

and then

that vector

set gets

fit in a way that's more tractable than

just this sort of endless sampling oh

what if there was a whole body and you

just missed it type questions however if

there was a whole body in your library

or the family of functions you were

trying to fit were only whole body

then uh

i mean

you can have a mismatch there just like

you could fit any other model

incorrectly

so the markov blanket is one of the um

terms that i think many people are

curious about certainly

ourselves included

and

from

number 14 livestream number 14
we talked about this continuum slash
development of the concept from
the way it was phrased by
markov by the way we're still trying to
figure out which one
uh
slash exactly which contribution the
father and the son made
towards the development in pearl 1988
in the
uh textbook
uh
shown here
and this is where computational methods
start to come in
and some of the pure math assumptions
get deployed in real data sets so then
the markov blanket is defined as the set
in a bayesian graph of insulating nodes
so moving from this matrix
representation towards this graphical
bayesian representation
and then
there's some of the developments by carl
fishton and others which this paper is
part of
that do a few different things relating
to connecting
separating the
blanketing states
into incoming sensory and outgoing
action
so partitioning of the blanket
as well as
implementing some of these components of
principle of least action which is not
implicit in just mere statistical

insulation on a bayesian graph so we'll
talk about some of that stuff but i
think there's enough to cover in what
this paper
here adds and
number 14 number 20 number 26
many other papers
that we've discussed in many that we
haven't have very insightful
uses as well as criticisms of markov
blanket concept
so what is the free energy principle
this
keyword one way to answer it since it's
a
keyword in many papers
what are three possible attentional
regimes that you might be able to expect
will reduce your uncertainty about this
question because if you're not worried
about this question
it's all good then but if you are you
want to reduce your uncertainty so what
are the actions to reduce your
uncertainty
well the third sentence of this paper
that we're discussing here is
in brief the free energy principle is a
variational principle of stationary
action
applied to a particular partition of
states
where this partition rests upon
conditional
independencies
which is the other side of the coin of
uh
which which things influence each other

and which don't kind of two sides of the coin there

there's then a link to citation 15.

so citation 15 if you want to go one more layer into

establishing whether this is a mereditus claim or not

there's friston's 2019

monograph free energy principle for a particular physics particular being specific but also referring to particular or autonomous states as they're known

and also much discussion has occurred since this paper so

look forward to seeing how

it develops

and then a third possible attentional regime would be to come participate and contribute with actimflab because every day

we are reducing our uncertainty about terms as well as any participant can attest

and we're always trying to improve our informational niche through projects like educational courses and ontology development so that the questions that people ask

every day like what is a markov blanket why does it matter what is the fep why does it matter

there will be awesome resources to share about those questions

one other piece before we jump into the paper is just a little bit of matrix math vocabulary so i hope that i'm

just even quoting the definitions
correctly because i'm not super familiar
with a lot of these distinctions
and would think that there's a lot more
expertise that people could share which
would be awesome so the jacobian matrix
and the determinant
is a construct in vector calculus
and what's relevant here is that it's
the matrix of the first order partial
derivatives so if you have just like a
lump in space you can take the partial
derivative it's like putting a ruler and
finding the derivative of a certain axis
of a certain variable
at that spot
and so you can find the partial
derivatives of higher order
models
the hessian matrix
is defined as the square matrix of
second order partial derivatives
of a function or field
it describes the local curvature of a
function of many variables
so same idea it's like
how
the the linear slope and then also the
curvature so that can tell you a lot
in
especially
a certain case
that's going to come into play in how
this paper
defines its analysis approach
specifically the laplacian approximation
so the abstract
in this treatment of random dynamical

systems we consider the existence and
identification of conditional
independencies at non-equilibrium steady
state

these independencies underwrite a
particular partition of states in which
internal states are statistically
secluded from external states by blanket
states

the existence of such partitions has
interesting

implications for the information
geometry of internal states

in brief this geometry can be read as a
physics of sentience where internal
states look as if they are inferring
external states

however the existence of such partitions
and the functional form of the
underlying densities have yet to be
established

here

using the lorenz system as the basis of
stochastic chaos we leverage the
helmholtz decomposition and polynomial
expansions to parameterize the steady
state density in terms of surprisal or
self-information we then show how markov
blankets can be identified using the
accompanying hessian to characterize the
coupling between internal and external
states in terms of a generalized
synchrony or synchronization of chaos
we conclude by suggesting that this kind
of synchronization may provide a
mathematical basis for an elemental form
of

autonomous or active

sentence in biology

okay cool

how will that happen here's the road map

the first section is the introduction

the second section

from dynamics to densities makes the

connection between dynamical models

and

densities

which takes one towards flow like if you

have water in the ocean of different

temperature and densities you can do the

flow modeling

section three

is the helmholtz decomposition which

we've also revisited elsewhere

and it relates to

splitting apart a vector field into

two components one that's kind of like

hill climbing and one that's more like

curvature

the lorenz system a classic

complexity

model

is brought up

and

used as an initial case

for this kind of dynamics to densities

plus helmholtz decomposition

then some further

lorenz

systems other than the lorenz are

analyzed and then the laplace and

systems other than the laplace are

analyzed

section four markov blankets and the

free energy principle

talks about sparsely coupled systems and

specifically where the different partitions are engaging in some type of synchrony which is going to be like correlated action conditioned upon the blanketing states that leads to a discussion of particular partitions boundaries and blankets and a kind of revisiting using a simulation analysis of markov blankets the last section is about the free energy principle and mentions active inference so it's pretty interesting just to start with the data availability statement they wrote that the code used to run the simulations and create the figures for the paper are freely available in the dem toolbox of the matlab package spm12 so if anyone maybe wants to like do spm live with us that who knows how to do it that'd be awesome spm stream it's be great so let's start with the introduction slowly because there's going to be some more technical parts later the last sentence of the first paragraph of the introduction is from this one can elaborate a physics of sentience or bayesian mechanics that would be recognized in theoretical neuroscience and biology so imagine if the top half were redacted what do you think would allow for quote the elaboration of a physics of

sentience or a bayesian mechanics

so

not possible

or it's already been done who did it or

well it would have to have this

attribute it'd have to be able to run on

this type of computer or it'd have to

have this many variables or fewer so

what are we looking for here

what would justify

this sentence

that they start the paper with

and why would it matter which is kind of

tied up with maybe what would um be the

motivation for developing it however

much effort

it

involved why would it matter to have a

physics of sentience well

why did it matter to have a physics of

heat

let's pull back one more sentence

the internal states can then be cast as

representing in a probabilistic fashion

external states

so that's what they're going to

elaborate a physics of sentience on from

this partitioning of internal and

external states

but not just any partitioning

a particular partitioning where internal

states can be cast as representing

external states

so this is kind of a minimal

representation we have internal and

external states and we're not going to

define them as even action or sense

partitions in the blanket yet just

there's some partition between internal and external states and the question is is this how you think the world is partitioned or how our statistical models are positioned what is this partition and we've talked about that in a few different papers and that's going to be a big part of this paper then we can pull back one more sentences in the abstract or i mean in this first paragraph where they define the free energy principle which were which we read previously and then specifically if the states of a system whose dynamics can be described with random or stochastic differential equation eg the length vine equation possess a markov blanket then an interesting interpretation of their dynamics emerges the conditional independence in question means that a set of internal states are independent of another external states another external set when conditioned upon blanket states and so that is the blue here's the precondition for the red claim which is going to be what is the basis for in the author's words elaborating a physics or sentence so that is kind of where we're going in this paper agree or disagree there's going to be a

lot of specific points where juan can
raise a qualm and that's totally okay
and i think a lot of the
discussion i hope will be very
interesting with people
seeing which parts of this presentation
resonate with them or their
understanding or not
those are the big questions what kind of
blankets line up with different parts
aspects or partitionings of the world
map is not the territory etc
what makes some partitionings in the
world act informationally like
inferentially anticipatory
predictively etc
why are some partitions able to be cpus
and other partitions are just
divided halves of a cup of water
how do partitionings of
variables that model the world
lead to a physics of sentience and
action
is there another way to get to a physics
of sentience and action
is there another way to do physics
they
describe
their approach to part one
by
applying the helmholtz decomposition
which
we're going to return to so this is just
kind of a a preamble and we'll talk more
about some of the keywords but
they're going to apply the helmholtz
decomposition to the solution of the
focker plank equation describing the

density dynamics of any random system

and then

later on they're still summarizing part

one and they're going to say we

approximate the flow

from the yellow part

with a quadratic expansion which means

the steady state density reduces to a

gaussian form

so it's kind of like if you fit a

quadratic model

$ax^2 + bx + c$

to

some

data set

it's not a statement that the data set

is going to have a derivative that's

linear and then a derivative of that

that's a constant

that's not

part of the data set of the world that's

about the quadratic model which is a

very constrained model set that you

chose to fit

and maybe other models would have fit it

better other ones would have had more or

fewer parameters but you knew something

about the form of the solution because

you constrained it to a certain set of

parameterized equations which is why

they talk about variational inference

and then

they're going to show that for a lorenz

system

subject to appropriate random

fluctuation

there's going to be this third state

which is independent of the first two

states or dimensions and this means that
the third state has no blanket states
and therefore is not is no there's no
particular partition of interest

so

one question for anyone who maybe
understands the formalism just why does
this claim specifically

matter

so section two

the aim of this section is to get from
the specification of any of a random
dynamical systems in terms of its
equations of motion to the probability
density over its states in the long term
future from any initial conditions
we are only interested in systems that
have a limit set

in other words

systems that possess an attractor
we want to describe systems that do not
necessarily visit all possible states
are not necessarily time reversible
are stochastic in nature
and potentially chaotic

in short we are interested in stochastic
chaos in systems with a pullback
attractor and then there's several
citations

from the 90s and

maybe that's some of the earlier models
of it

that's

how they describe it and then they
show this formalism where the lang vine
the change in the state of x
that's kind of like a derivative

uh sign sometimes

is equivalent to like a flow and then
these faster fluctuations so kind of
like a wave and a ripple but that's
going to describe
the change overall of some location
and then that is
equivoc equivalated to
the focker plank density dynamics so
they're the same thing that's the equal
sign
so that
i can't say whether this is just a total
tautology super obvious or whether this
is a
conclusion that was only more recently
understood that would be helpful to no
but that's one of the baseline
equivalences in the paper is going to be
you can have
a specification of a random dynamical
system like here's the set of equations
that describe how this sensor is
returning data
and then
turn it into
a flow
and a density
estimation framework
so the plank
they write the dot product the dot
notation
indicates a partial temporal derivative
in the absence of random fluctuations
i.e $\gamma = 0$
the attractor corresponds to a limit set
namely the set of states onto which the
solutions of one converge so
uh there's kind of like

uh

some steady state

attractor

which

is when there's a heat differential like

there's a it's warmer outside it's

cooler inside and then

there's some

point where the thing

gets a zero

when the flow shows

exponential divergence of trajectories

with positively open of exponents i.e

real positive eigenvalues in the

jacobian of the flow the system can be

said to be chaotic

so they're systems that kind of converge

nicely when they're pulled away a little

bit so let's just say that it stayed the

same temperature inside and outside in

that previous example then that

equilibrium would be like pretty nice

because it would return to it if it

slightly

were moved away from it but there's also

systems

where nearby points

diverge and that's a measure of

this

approximated by this leopardive exponent

and then the question is for systems

that do have positive

leading leopardive

vectors

then

what happens to those

flow

models

and then what happens when
there's
a um
reinstating of the random fluctuations
how does that sort of system that would
be chaotic if it were allowed to just
somehow deterministically evolve
so it's like you're fitting the
underlying model and it's a little noisy
around it and then you swap out the
actual scattering of the real world
populations of deer and rabbit for just
the partial
derivatives that you were estimating
and then maybe you see that there's some
bifurcation in the population density
but of course any real population is
either going to experience one branch or
the other of that bifurcation
so
that's kind of the modeling that they're
working on
and then when you go from that dynamical
systems approximation to the flow
if the flow shows exponential divergence
of trajectories we can impute stochastic
chaos
so they take one of the classic systems
of deterministic chaos
uh
and dynamical chaos the lorenz system
which is kind of a butterfly looking
attractor
and they
um
uh first
show the trajectory of lorenz attractor
with and without this reinstating of

random fluctuations
so we know that there's a
divergence
Lyapunov divergence that's
making the system chaotic
when there's no random
fluctuations so what happens when that
is reintroduced
the middle panels
show the corresponding solutions in the
three-dimensional state space
illustrating the butterfly shape of the
limit set
so
even when the μ
and the random attractor so the
deterministic one is the kind of clean
butterfly
and then the μ
uh one with a random fluctuation
the points are outside of
the butterfly it's a slightly different
shape
but then it still illustrates it's
not a uh
you know
an x
in the state space so it still
is kind of following this but just
it's of course it's not
following the exact tracks that it's on
because there's a random fluctuation
term but it still is going to have a
certain
geometry
because this is a model hence
information geometry
so

that's going to get some transformations
and approximations
lower left panel plots the fluctuations
in the potential evaluated using the
laplacian form this is kind of formalism
that's going to come later but expresses
the self-information
as an analytic
function of the states
and then
the states
i was not exactly sure why they were
called states rather than dimensions so
if anyone has a thought on that that'd
be interesting and then the lower right
is the potential function of the first
two states basically or dimensions
is shown as an image with the trajectory
superimposed
so this is kind of the um
even this smoothly varying global
potential
plot
with the
lighter and the darker representing
different values there's this like
dynamical butterfly
that's
fit from the re-sampling
of the true lorenz
all right so the key insight from figure
1 is that the underlying system this is
kind of a key complexity
insight that the underlying system can
have very simple and defined rules yet
unpredictable dynamics
so in the case of dynamical equations
it's systems like the lorenz attractor

and many other
systems that are just very simple and
they only have a few parts yet they
result in these very incredible
and
rich
behavior
and then there's also kind of like
simple rules complex outcomes you know
ants and humans so just a simple rule
can be underlying something that's
giving
an extraordinarily complex pattern
and
yet if we measure these systems
empirically
that um have some simple rules
underlying them but there's also this
random fluctuation or we don't sample at
all
there will be some but not overwhelming
noise there will still be dynamical
patterns to pull out
so
we can still reduce our uncertainty
about the location of future points by
using a statistical model that we
defined that doesn't even have to be
similar to the generative process
so if someone said well you get money
for getting
predictions closer to the actual
position of the sampling
even if your model that you're fitting
to it is like a neural network or is
this kind of model that we're going to
discuss here rather than a differential
equation that's actually the true

generative model
you can still
do better than just guessing within this
cube
so section 3 goes to the helmholtz
decomposition
and they write u_m
so that was the previous section was we
can regard the deterministic lorenz
system as describing the expected flow
of a random dynamical system that is
subject to random fluctuation that was
what we just talked about the flow of
such systems which possess a
non-equilibrium steady state density can
be expressed using a generalization of
the helmholtz decomposition into
dissipative which is irrotational and
curl free and conservative which is
rotational divergence-free components
where the latter are referred to as
solenoidal flow
for an introduction to the
generalization of this helmholtz
decomposition see appendix b of 16 which
is
bayesian mechanics stream number 26.
so
we're looking for where p
dot so the partials
of p partial derivatives of p are zero
so where is
that
flow equilibrium
well
that is going to
be a function f
that has a few

terms including one that's
i think introduced in this paper the
housekeeping function
but it has two terms that are the
classic helmholtz decomposition which is
the solenoidal the tangential current
that's just the spire the iso contour
and the gradient which is kind of like
the up and down
so that's how a total current in
electromagnetics gets decomposed and
it's related to um kind of fundamental
vector field math
so
let's go and uncover what was hidden by
this gray box
so we have the the focker plank steady
state p of x equals zero p dot of x is
zero
there's some function that's going to be
a decomposition of that into
a solenoidal and a gradient flow
component the helmholtz decomposition
and then
this housekeeping term
which is an append in appendix but um
didn't go into in super depth
so curious about what that is but it
results in this whole flow term
here is the fancy eye
maybe it's a fancy j the first
dissipative part performs a romanian
gram gradient descent on the negative
logarithm of the steady state density
which can be interpreted as the
self-information
which is the
negative

natural log of p of any given state
or as some potential function
so that's kind of like how surprising is
that state the second part of the flow
is the solenoidal circulation on the
isocontours of the steady-state density
and then the third term is this
housekeeping term that accounts for how
changes in the flow operator
change
so it's a um
extension of the helmholtz decomposition
thinking about how to solve dynamical
equations
so this is from active26 the
synchronization map and
i'm not going to go into the whole thing
because you can check out 26 but the key
question is about this sigma this
synchronization manifold
between the internal the external states
conditioned on the blanket states
so
how do you insulate yet have
anticipation across that installation
and then also from 26 this is kind of
what it looks like to see a stochastic
trajectory
decomposed into a gradient
and a rotational component
all right 3.1 our objective is to
identify the functional forms form of
the self-information or potential
function that describes the
non-equilibrium steady-state density
so why
that's
how surprised you should be

that's the potential function

that's sort of the um

[Music]

the fund the fundamental frequency of

both the solenoidal and the gradient

so it's an important variable to know

but how is self-information measured or
calculated

how does the self-information for
different non-equilibrium steady-state
or equilibrium study states compare

what does it mean for self-information
to be high or low

so to be surprised or not

so first

they um in formalism 4 address this set
of questions basically for one

simple er case which i'm not super
familiar with the matrix

math so we'll leave it to others to
evaluate some of the technical details
so then they

follow up and say basically

we could solve that easily without that
correction term

that arises when flow operators are a
function of the states indeed it is this
state dependency that underwrites
stochastic chaos so that's like the
double pendulum

the reason why it's a chaotic system is
because the states at the time depend on
the previous time

in a way that generates chaos

so there's so much dependence over short
term that things can diverge really fast

this present this presents us with a
more difficult problem the problem can

be finessed by using polynomial expansions of the flow operator and the potential as follows for n states up to polynomial order m so they're going to use polynomial expansion approximation on a complex underlying flow functional polynomial expansions restrict and scaffold our model selection approach to a manageable size and tractable computation so it's kind of like the Taylor series and the Volterra series and other polynomial expansions which can fit functions often that are very complex well and so this is showing you the Taylor approximation to a sine wave and showing how the first approximation gets you like between 1 and negative 1 the red line might be totally more than enough but with only a few more terms being added all of a sudden it starts fitting really well further and further out so that doesn't mean it can go forever but the polynomial expansions can be very powerful in the numerical case like just doing good enough with quick and dirty as well as in the analytical case here's a figure from SPM 2007 textbook which is a great textbook by the way and it's talking about the Volterra series as a general non-linear input state output

characterization system so

nb no to bene good note to you

volterra kernels are synonymous with

effective connectivity

and then they write

this is from the spm

12 so later than this but right up from

the point of view of regression models

modulatory effects can be modeled with

nonlinear input output models and in

particular the volterra formulation

described above

because the kernels are high order they

embody interactions over time and among

inputs and can be thought of as explicit

measures of effective connectivity

an important thing about the volterra

formulation is that it has a high face

validity and biological plausibility

the only thing it assumes

is that the response of a region is some

analytic non-linear function of the

inputs over the recent past

this function exists even for many

complicated dynamical systems with many

unobservable state variables

so

cool

why does it matter to do a polynomial

expansion of the functional form of the

solution to this flow problem

the parameterization in 5 allows for

state dependent changes in the amplitude

of random fluctuations

encoded by the leading diagonal of the

flow operator

with that functional form of the

polynomial it is straightforward to

solve for polynomial coefficients the
flow operator by solving the following
simultaneous equations for a series of
sample points
some of the keywords that are
from spm and elsewhere that might come
into play to learn more about these
topics are expectation maximization
optimization
parametric empirical base
dynamical systems identification and
generalized linearization schemes for
fitting
glms to
non-linear systems
and then something that's not as much in
spm
but comes into play later with active
inference and hopefully we'll be able to
demonstrate this in the literature as we
sort of sift through it
is that action planning
amidst uncertainty and feedback with the
environment
radically changes the nature of that
kind of flow estimation problem
you can't
um use the same model necessarily for
the
um
leaf on the stream as something like the
winged snowflake that's actively
resisting
so
how good is this approximation
to use the polynomial expansion
of the fitting of the random
fluctuations of the random dynamical

system that is chaotic
inspection of the lorenz system suggests
that a second order polynomial
approximation is sufficient given the
flow of second order in the states
this unsought so
vocab
onsots
an assumption about the form of an
unknown function which is made in order
to facilitate solution of an equation or
other problem
has an interesting implication if the
self-information can be approximated
with a second order polynomial it means
the non-equilibrium steady state is
approximately gaussian this is known as
the laplace approximation in statistics
here we generalize the notion of a
laplace approximation to cover not just
the quadratic form of the log density
but also the solenoidal flow that
underwrites non-equilibrium dynamics
big if true
it has to be
so
here's the helmholtz decomposition
so now we're taking
that
solution that we had earlier
uh and thinking about it in terms of
fitting only up to the second order
polynomial expansion of the flow
solution so that is as they say what is
known as the laplace approximation in
statistics and it's in spm textbook 2.
here's the actual flow decomposition of
the lorenz system

the solenoidal flow is red the gradient flows blue and the correction term is gold

the flow is shown as a quiver plot at equally spaced points so it's kind of like the vectors of different direction at different points and then the right is the same but for the laplacian based upon the one on the left

here

the key difference is that the dissipative part of the flow operator and hessian are positive definite indicates the laplacian which means the gradient flows converge to the maximum of the non-equilibrium steady state density this is reflected in the blue arrows that point to the center of the space

so even though the gradient of the real flow is pointing all the different directions because it's kind of like a mixing flow so there's all the vectors are going in different directions

but then in the laplacian second order polynomial expansion the blue arrows are all pointing to the center

or to some manifold that's much more compact so it's kind of like a converging solution even though the actual system we defined it to have stochastic chaotic properties so

here's spm

looking at the lorenz tractor

alas poor lorenz

i knew it horatio

a model of infinite citations of most
excellent fancy it hath returned to the
pullback attractor a thousand times
and now how about in my imagination it
is

so that's

modified

from hamlet

because right after that seemingly very
cool

solution

they say

indeed the equations of motions
we have written down are deterministic
implying that there is no stochasticity
in the sample paths of the process
assuming a fixed initial condition this
speaks to the fact that helmholtz
decomposition of the deterministic
lorenz system is not a description of a
dynamical system with random
fluctuations ie systems in which the
dissipative part of a flow operator is
positive definite

we therefore need to look beyond the
lorenz attractor so that takes us to
appropriately 3.3 for beyond the lorenz
system so okay cool

now beyond lorenz system we go
to the chaotic laplacian so figure three
this is um a trajectory kind of like a
pointless trajectory of three states
kind of three dimensions comprising a
laplacian approximation

to a stochastic lorentz system so here
it's second by third

first by third first by second of the

nest density so it's showing three
projections like the geb cover
onto a two dimensional and notice how
the light is in the center so it's like
they're getting pulled back
to the peak of the gaussian
on each dimension
so that's the non-equilibrium steady
state which by construction in the
laplacian is a multivariate gaussian
that's super key and extremely
interesting
the middle panels show deterministic and
stochastic solutions as a function of
time
while the right panel plots the same
trajectories in the state space the
shape of the attractor retains a
butterfly-like form but is clearly
different from the lorenz attractor so
what do you see when you look at this
attractor
be honest
but it has a butterfly-like shape and
the um
generated trajectory has some
uh similarity with
the
underlying generator function but it has
a such a nicer form to solve
and then the lower left is the self
information the potential as a function
of time based upon the analytic form for
the equations of motion
and the deterministic trajectories of
the previous panel
and then
the flow of the laplacian which is the

approximated
expansion
nicely solvable etc
against the lorenz
that's the true flow is evaluated at 64
space points it can be seen that
although there is a high correlation
between the flow of the laplacian loren
systems they are not identical so not
super visible could have provided like
an r squared or some other information
but clearly they are um drawing from the
same manifold
so
if this is good enough to stay alive
then
c'est la vie
upper panels just focus on those so
this is again the projection onto two by
three one by three and one by two
of three
of this multivariate gaussian
laplacian approximation which makes it
really nicely solvable so one can then
estimate the lyapunov dimension
which is approximating to the house door
hausdorff dimension
which in this example was 2.48
so
the laplacian approximation
was used to sample
and then estimate the lyapunov dimension
and then it turns out that
that compares very closely 2.43 and 2.48
which in the grand scale of things where
above one means chaotic system
it means that the laplacian
expansion has some of the nicely

solvable characteristics of the
generalized laplacian like the
multivariate gaussian and the easy
computation
yet it also identifies the system as
chaotic
because it's it still gives a positively
open exponent
from this sampling
in short the flows of the loren system
and its laplace approximation have an
attracting set with between two and
three dimensions for our purposes the
laplace approximation is easier to
handle than the lorenz system
because the functional forms the flow
and potential are immediately at hand
so

pretty interesting

figure four

so now

this recalls our earlier discussion of
the jacobian and the hessian

so

let us recall the jacobian is the first
order partial derivatives

the hessian is the second order partial
derivatives

okay

so

we have the jacobian

the hessian and the covariance

between these three dimensions

so

the lorenz attractor was in three

dimensions okay it was in state one two

and three that's kind of why there's

it's like thermodynamic states so there

are dimensions though so
the lens attractor was this like kind of
you know ant flying around in space in
three dimensions
so we fit a polynomial expansion
specifically to those three dimensions
the measurements of them
now for those three dimensions
we can talk about
their correlations
in the first partial derivative
and the second partial derivative and
the covariance
that's in the
top here
in this example the third state is
independent of the first pair
where the independence
rests on the directed coupling from the
third to the first state
and so it's logged to show the sparsity
so
do you see where we're going with the
markov blanket
see how this third state
it's not part of the lorenz system
underlying differential equations
it's actually just a total figurement of
the laplace approximation model
but it's quite an interesting pattern
the middle panel
shows slices through the steady state
density over the two states
um
[Music]
at increasing values of the remaining
state
so the only correlation in play is

between the first and second state so
this is related to
sphericity error correction and
evaluation
and it's a topic that is discussed in
spm textbook
but basically spherical errors are the
ones where the two axes are not
correlated but then when there's
non-spherical errors then there's
you get a different shape
the lower panel the correlation
is illustrated in terms of the
conditional density over the first state
given the second
so this is kind of showing that
by and large
it's
doing well
largely confined to the 90 credible
intervals would be good to see some you
know
simulation statistics and raw data on
that
so
again alas
i knew him well
the preceding treatment leverages the
simplicity of the laplace approximation
to stochastic chaos in which sparsity
constraints on the hessian are easy to
identify or implement
now
just like lorenz was dropped
now this is going to be
going beyond the laplace
system
so

i'm going to go into too much detail
here but that's
that's a section that people could
explain more about or we could look at
more in the dot one the dot two so three
point five summary
it is important to not conflate the
simplicity of a ness density with the
complexity of the underlying density
dynamics
in other words when prepared or observed
in some initial state the probability
density can evolve in a complicated and
itinerant fashion on various sub
manifolds of the pullback attractor so
the map is not the territory that's
something that has come up many times
the laplacian is not the lorenz
but
the laplacian is a good approximation to
the lorenz
in their simulations
figure six
so
now
the chaotic laplacian system
is um
it's kind of shown in a time series way
where this is the true
relationship
between um the first and the second
state so let's go to here so we see the
first and the second state
here's like where there's this
non-spherical error
and so this is kind of like an
interesting like coupling in the model
that's independent of the random

fluctuations

and so that represents something very
deep about the structure of
the laplacian approximation to the
lorenz

now

using the laplacian in a generative
capacity

there's this initial misattunement in
their setup it starts somewhere far from
the pullback attractor

and then like the correlation evolves
through time and then they say

the density converges to the steady
state density after about 16 seconds

however it takes a rather circuitous
route from this particular set of states

okay

so that's interesting it starts with

like they were differently correlated

because of they starting in a weird

place and then they found their true

correlation structure note that the

average density over short periods of

time can be highly non-gaussian

even though the density at any point is

by construction gaussian

something to think about

so then we get to finally markov

blankets

so they repeat the analysis but now they

approximate two lorenz systems that are

coupled to each other through their

respective first states so the first and

the second states were coupled

so it's almost like one of our coupled

states we're gonna

couple out now

this induces a richer conditional
independent structure from which one can
identify internal and external states
that are independent when conditioned
upon blanket states so

what does it mean to sparsely couple
systems

and what are these quote first states
the first dimension okay figure seven so
now there's generalized synchrony in a
laplacian system

so this is now the same figure as figure
3 like this

except now we're looking at
the generalized synchrony
so now there's a synchronization
manifold in

the first
state when those two got linked up so
now there's more lines because there's
like two coupled chaotic systems
but the middle is showing that their
state spaces are
coupled

and this is illustrating the degree of
synchronization
and then again showing that their flows
are correlated no stats

figure eight
now instead of having

3
by 3 in the matrix correlation and
covariance in the first and the second
derivatives now it's
6 by 6.

so here is now the identification of
using the log of the jacobian
and the hessian to identify

different partitions
so that partitioning between the one and
the four
between that first state of both
is the one that we knew about that's
like
the the one that was wired into the
system
but then the other ones are showing
other correlation patterns now for i
don't know why this four and five
connection is not included in a box but
these are the partitionings
and so um
the sparsity structure of the covariance
supports a particular partition into
internal states five and six dark blue
active
fourth state the one that induces
the one that connects
here
sensory first state and external second
and third state this partition is
illustrated here and the remarkable
thing is that despite their conditional
independence there are correlations
between internal and external states and
here between the second and fifth states
so
interesting patterns
showing these evolving through time
nine
so now they're going to look at the
partial correlation of the states
um and showing that the hessian
is
recovered
so here is like that

hessian on the top right
and then now they're gonna show
that it's recovered it's a little bit
light
but it's showing how
these so so these squares
are showing the partial correlation
coefficients and hopefully pulling out
the similar structure as identified here
and then um like the strong correlation
well
yeah we'll investigate that more in the
dot one the dot two and then showing how
like the third and the sixth state
for example so that's like three and six
they're not supposed to be coupled
and so then they initially start off
with a high correlation because maybe
they start in the same neighborhood but
then they converge towards a zero
correlation
so
that is
sparse coupling
it does not depend upon gaussian
assumptions about the non-equilibrium
steady state density it implies that the
dynamical influence graphs with absent
or directed edges admit a mark called
blanket which may or may not be empty
these independencies can be used to
build a particular partition using
following rules
capture the flag figure 10.
so here's where we see kind of the
classic bacteria example but now it's
two bacteria communicating i'm not sure
if it was always that way and then here

is a little bit more
of a coupled systems with two times
three
which is pretty interesting
so here is where we see the mark all
blanket as we've seen it before but also
in a new way
how great
here's some technical definitions of the
markov blanket so
not for tonight but
for another night
so um
here's a question they asked themselves
why why one might ask why does a
particular partition comprise four sets
of states
i don't know why is the tetrahedra the
smallest polyhedra who knows
in other words
why does a particular partition consider
two markov boundaries sensory and active
states so why can we skip straight to
the fristen blanket rather than just
identifying the pearl 1988 or the markov
earlier model
the reason is that the particular
partition is the minimal partition that
allows for directed coupling with
blanket states
so that again is from figure 10
and the key thing to note from figure 8
is that there are profound covariances
between some internal and external
states despite the fact they are
conditionally independent
so
we're defining like nodes that are not

connected are conditionally independent
that's how the bayesian graph works
so here one and four get correlated
get a connection with each other that's
this connection right here right there
and right there
so now they're coupled
one and four so even though
some of the internal states
are so now it's like
the external states are
at minimum two and bucky fuller
famous quote unity is plural at minimum
two the internal states have to be at
minimum two
external states have to be at minimum
two
so that's pretty interesting
because they yeah
um
yeah
the particular partition is the minimal
partition that allows for directed
coupling with blanket states
okay
and then um
they're the second the fifth states are
highly correlated yet are
twice removed
and so this is leading to a general
synchrony through partial coupling
of
dynamical systems
note this is their words note there is
no claim that either the original lorenz
system or coupled lorenz system
possesses a markov blanket
the claim here is that there exists a

laplace approximation to these kinds of
systems that in virtue of the zero
elements of the hessian
feature markov blankets so
oh how many days
the having a markov being a markov
modeling a markov
no claim
a laplace approximation
approach to systems identification can
add constraints and make the problem
easier to solve
the markov blanket is arising
literally within the laplace
approximation
which is just an approximation it's not
a description of the system
so
on to the free energy principle
the existence of this any particular
partition means that one can provide a
stipulative definition of the
conditional density over external states
being parameterized by the conditional
expectation of internal states given
sensory states
we call this a variational density
parameterized by expected internal
states μ
so
going from
the minimal coupled systems
to
there are still some persistent
covariances
and these covariances
partition out
into

a markov blanket

what if

the expectation of the internal states
were of external states

in other words or in words for every
sensory state there's a conditional but
it has to be conditioned on the blanket
state which is how the whole thing is
set up in words for every sensory state
there's a conditional density over
external states and conditional density
over internal states where internal and
external states are conditionally
independent

this admits the possibility of a
diffeomorphic map between the sufficient
statistics of the respective densities
the existence of this mapping rests upon
a continuously differentiable and
invertible map

which is linear under a laplace
approximation so diffeomorphic is like
stretchable

so that allows

some of these formalisms to arise
again would be awesome to hear what
other people

can bring to the table about this but
this is like the energy minus entropy
form the divergence term plus the
self-information about the policy and
then the accuracy minus the complexity
so would be awesome to learn more about
this

okay the font got a little out of
control here but
this functional
the big one

is going to be expressed in a few
different forms
so it's an expected energy minus the
entropy
that's energy minus entropy
self-information
plus kl divergence okay self-information
plus kl
negative log likelihood of particular
states and the kl diversions accuracy
minus complexity okay accuracy minus
complexity
it's also in the machine learning
context
evidence lower bound elbow
this is the basis of the free energy
principle
put simply it means that the expected
internal states of a particular
partition at non-equilibrium steady
state can be cast as encoding
conditional or bayesian beliefs about
external states
equivalently equivalently the flow in
the internal manifold can be expressed
as a gradient flow on a
variational free energy
that can be read as self-information
this license is a somewhat teleological
and directed description of
self-organization as self-evidencing
in the sense that the surprisal or
self-information that constitutes the
potential is known as log model evidence
or marginal likelihood in bayesian
statistics so
wow
great fep

work

what does it mean to talk about the
physiology perspective here though so
what does it mean for self-evidencing in
the biological context the blood vessels
wanna observe themselves working
so now in section five one getting
hopefully towards the end here
um

in alternative this is later in section
five one in alternative and deflationary
perspective

rest on noting that free energy
gradients are also the gradients of
self-information so the organism wants
to be minimally surprised about
some chaotic
pullback attractor

so whether or not things are wacky out
there or not
having this updating flow
model

that comes to a pullback attractor
that is of the approximation that the
modeler controls is very important and
the organism wants to be minimally
surprised about the expectations and
preferences that it has over
observations and that is defined with
some technical details here

so

this formulation of gradient flows is
simpler and shows that they're
effectively minimizing a different sort
of prediction error
namely the difference between particular
states and the expected values at
non-equilibrium steady states

this leads to exactly the same gradient flows but a complementary interpretation in which autonomous states are drawn towards their steady state expectation heuristically one could imagine this kind of stochastic chaos as apt to describe the motion of a moth attracted towards a flame but constantly being thwarted by turbulent i.e solenoidal air currents

because active states influence sensory states and possibly external states this would look as if the particle eg the moth was trying to attain its most likely state in the face of random fluctuations and solenoidal dynamics this perspective emphasizes the active part of self-evidencing sometimes referred to as

active inference

so how cool first in 2006 the winged snowflake

21

like a moth to the flame

5.2 so they um

uh

give more

summarization here about that sequence that they took of constructing the laplacian approximation with a quadratic form and then looking at that first and second

partial differential uh partial

derivatives matrices the jacobian and the hessian

looking at the conditional

independencies then the generalized synchronization synchronizing

given the um partitionings
and then
suggesting that there's some
diphyomorphic like stretchable
connection between the internal and the
external states conditioned upon all of
that and then that's the bayesian
mechanics
in which internal states on average
parameterized beliefs of a bayesian sort
about external states

okay

so

some of the implications in the last few
slides here so clearly the worked
example based upon sparsely coupled
lorenz systems does not mean that a
conditional synchronization map exists
or is indeed invertible which would
allow the tail of two densities the
recognition and the generative model
in any given system

so you can't just specify any system
potentially and find a conditional
synchronization map

however the above derivations can be
taken as existence of proof that such
manifolds and accompanying variational
free energy formulation can emerge from
sufficiently sparse coupling

so

this is one system classic system of
complexity analysis and it worked here
maybe there's other systems where it
will or won't

a particular partition is necessary to
talk about states that are internal to
some particle or person

and which can be distinguished from external states one way of understanding the ensuing coupling between internal and external states is in terms of generalized synchrony that's that synchronization manifold this synchronization can be expressed as a variational principle of least action using the notation notion of variational free energy so this is from another decosta paper the synthesis on discrete state spaces and it is uh talking about how that variational free energy minimization of perception and planning as inference about observations can be seen as a generalization here's active inference um if you have no prior then you're you have surprise pure surprise optimal surprise intrinsic motivation info gain info max principle because it's just about what's surprising given no priors when there's no ambiguity so when there is a model that abstracts away from for example measurement error you have risk sensitive policies and kl divergence based control and occam's principle and there's other cases like bayesian decision theory expected utility theory as well as maximum entropy and jane so it's kind of um there's a lot of pieces that are coming into play when we talk about active inference probably some that will still be yet to learn

[Music]

but there are some pretty interesting patterns arising from this partitioning here's another form another implication so the polynomial form of the helmholtz decomposition from formalism 5 may provide a generic model for observe random dynamical systems in other words it could be the basis of a forward or generative model that explains some empirically observed flow estimated using the first and second moments to quantify the flow and covariance of random fluctuations over state space respectively this kind of generative model is appealing because of its parameterization in terms of the underlying non-equilibrium steady state density in other words one could in principle try to explain empirical data in terms of a gaussian steady state density which may include constraints on conditional dependencies so let's think back to the beginning with flow and information flow and mental flow and grain it um moving down a quarry or all these kinds of flow and then could there be something that integrates across them another option for scaling this approach to high dimensional dynamical systems would be to learn the state dependency of the flow operator and the non-equilibrium steady state density using unsupervised learning approaches such as deep neural networks

a similar approach has become popular in the deep learning literature in the form of neural stochastic differential equations

so here's two recent stochastic differential equation papers and then

they bring up this very interesting suggestion

one could use a separate feed-forward neural network to parameterize the different components of the flow operator and the self-information this would require differentiable transforms to be applied to the output layers of the network to constrain the hessian and solenoidal flow operator to be positive definite and antisymmetric respectively so

suggesting

some neural network architectures that might

facilitate

active inference

then this is a pretty nice

piece it is interesting to relate flows across a markov boundary to the constructal law

which like the free energy principle seeks a normative account of the structure and dynamics of complex systems

in the case of the constructor law this is articulated in terms of maximizing external access

to the currents internal to the system

so

here is um adrian bayan

who's like a awesome professor
and researcher who has done many years
of work
like fristen in some ways
um on this construct the law so i was
quite
happy to see that connection
here's another final implication one
practical implication finally the
practical implication great
it licenses the use of inverse sample
covariance matrices of a sufficiently
long time series as petition potentially
sufficient description of the steady
state density
so we could take a real time series and
fit that laplacian approximation
this would furnish a description of the
expected flow and leopard of exponents
to establish whether the system was
chaotic or not
in conclusion
last sentence
having a simple functional form for the
flow of random dynamical systems may be
useful for both modeling
and analyzing time series generated by
real-world processes that are far from
equilibrium that may or may not be
chaotic
so we can think of two chaotic systems
the humans and the ants and then
everything else
so
pretty interesting paper
um
the coming two weeks hopefully we'll
have some good group discussions we're

going to email the authors now that
we've done our work on the dot zero and
would be awesome to have them on for the
two discussions we'll have or for any
other time

um

thanks everybody who was watching i hope
if you are watching to the end you found
this interesting and
come get involved with active lab
because

we could all have some shared resources
so that some of these questions that are
really being advanced in this paper
like

the connections to complexity and
dynamical systems theory and
statistical thermodynamics um these are
important contributions and this is not
a fringe apologetics for active
inference it's barely mentioned in the
paper it's actually some really
fundamental work that

i hope

is understood and critiqued
where valid so what would a good
understanding here enable of the kinds
of things that are discussed in this
paper which came out in 2021 but think
into the future

what are the unique predictions and
implications of this paper what are the
next steps for fep and active inference
what are the goals of this research and
what are you still curious about

so

thanks for sticking around for a long
live stream at a different time but

so it goes

so have a good one everyone hopefully

see you around the lab or around

bye